

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Alterations in Cardiac and Systemic Metabolism in Heart Failure

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ABSTRACT: Altered cardiac and systemic metabolism is a hallmark of heart failure (HF). In the failing heart, cardiomyocytes develop alterations in substrate preference, mitochondrial oxidative metabolism, and the shuttling of high-energy phosphates from mitochondria to the cytosol that compromise energetic efficiency and contribute to disease progression. At the systemic level, neurohormonal activation plays a dominant role in HF with reduced ejection fraction, whereas HF with preserved ejection fraction is shaped by the clustering of multiple comorbidities, such as diabetes, obesity and hypertension, which disrupt the physiological crosstalk between the heart and metabolically active organs. This review provides a perspective on cardiac metabolism in HF. We delineate the specific alterations in substrate metabolism that characterize HF with reduced ejection fraction versus HF with preserved ejection fraction, examine the impact of interorgan communication on myocardial function, and highlight how the benefits of emerging HF therapies, including sodium-glucose cotransporter 2 inhibitors and GLP-1 receptor agonists, may be mediated, at least in part, through the restoration of metabolic homeostasis.

Key Words: cytosol ■ glucose ■ heart failure ■ obesity ■ sodium

The focus of heart failure (HF) pharmacotherapy is shifting from the heart to systemic alterations contributing to disease progression. Traditionally, HF with reduced ejection fraction (HFrEF) has been treated with neurohormonal inhibitors. However, several newer HF therapies exert their activity by primarily targeting extracardiac organs such as the kidneys, liver, and adipose tissue (AT), underscoring the importance of the systemic metabolic milieu in influencing HF outcomes. This paradigm is particularly relevant for HF with preserved ejection fraction (HFpEF), which is increasingly recognized as the cardiac manifestation of systemic disorders, including obesity, metabolic syndrome (MetS), and diabetes, rather than a disease driven primarily by intrinsic myocardial injury. Understanding how alterations in cardiac and systemic metabolism contribute to HF pathophysiology is therefore essential to fully appreciate the effects of contemporary therapies and develop new ones.

CARDIAC METABOLISM IN HF

Changes in Substrate Metabolism in HFrEF

In HFrEF, alterations in cardiac metabolism encompass substrate uptake, mitochondrial oxidative metabolism, and transfer of the ATP produced in mitochondria to sites of utilization through the creatine kinase shuttle (Figure 1). Furthermore, it is a long-held notion that the failing heart is metabolically inflexible; that is, it loses its ability to shift between different substrates according to nutrient availability and hormonal signals. However, a recent study in humans has challenged this notion, as increasing cardiac fatty acid (FA) uptake did not affect cardiac function or induce lactate efflux in patients with nonischemic HF.¹

Alterations in cardiac substrate utilization in animal models of left ventricular (LV) systolic dysfunction and in patients with HFrEF show considerable heterogeneity, explained at least in part by differences in the cause

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Nonstandard Abbreviations and Acronyms

β-OHB	β-hydroxybutyrate
AT	adipose tissue
BCAA	branched-chain amino acid
CaMKII	calmodulin-dependent protein kinase II
CKD	chronic kidney disease
EV	extracellular vesicle
FA	fatty acid
FFA	free fatty acid
GDF8	growth differentiation factor 8
GLP-1RA	glucagon-like peptide-1 receptor agonist
HF	heart failure
HFpEF	HF with preserved ejection fraction
HFrEF	HF with reduced ejection fraction
LV	left ventricular
MASLD	metabolic dysfunction-associated steatotic liver disease
MetS	metabolic syndrome
NAD	nicotinamide adenine dinucleotide
PANK1	pantothenate kinase 1
PCr	phosphocreatine
PDH	pyruvate dehydrogenase
SGLT2	sodium-glucose cotransporter 2
VAT	visceral adipose tissue

and severity of HF. One common metabolic derangement identified in the majority of studies is a shift in substrate uptake toward glucose and lactate and away from FA.² However, discrepancies have been noted in both animal and human investigations.^{3,4} The mechanisms underlying diminished FA utilization in HF are poorly understood. Although circulating levels of acylcarnitines are elevated in patients with HFrEF,⁵ analyses conducted by simultaneous arterial and coronary sinus blood sampling revealed that the human heart does not release acylcarnitines,⁶ and both FA and acylcarnitine levels are reduced in myocardial samples from patients with end-stage nonischemic dilated cardiomyopathy.^{7,8} Altogether, these findings suggest that the decline in FA utilization likely occurs upstream of FAO, at the level of FA import into cardiomyocytes.⁷

Several aspects of glucose utilization in HFrEF remain unresolved. While increased glucose uptake is observed in virtually all animal models of HFrEF,^{4,9,10} results in humans are conflicting, which could be explained by methodological factors such as fasting status or anesthesia, the small size and heterogeneity of study populations, and the frequent coexistence of comorbidities linked to insulin resistance.⁶ Importantly, increased glucose uptake is not invariably coupled to enhanced mitochondrial pyruvate oxidation. Alterations in the coupling

of glycolysis with pyruvate oxidation may reflect distinct stages of the disease, that is, compensated hypertrophy versus overt HF.^{10,11} In patients with end-stage dilated cardiomyopathy, myocardial levels of glycolytic intermediates are reduced compared with those in nonfailing hearts, possibly due to reduced glucose entry into glycolysis and increased influx of pyruvate into the tricarboxylic acid cycle, the latter favored by increased pyruvate dehydrogenase (PDH) activity.⁷ To meet the increased demand for pyruvate in the setting of impaired glycolytic flux, the failing heart increases lactate uptake as an alternative source of pyruvate (Figure 1).^{6,7}

Experimental models indicate that, particularly in pressure overload-induced hypertrophy, glucose is diverted toward anabolic pathways to support cardiomyocyte growth rather than ATP production, akin to the Warburg effect in cancer cells. Supporting this concept, increased glucose consumption via pyruvate conversion to aspartate sustains cardiac hypertrophy in mice after transverse aortic constriction.¹² In this context, the mitochondrial pyruvate carrier and PDH emerge as key regulatory nodes. In mice subjected to transverse aortic constriction, decreased mitochondrial pyruvate carrier activity limits pyruvate oxidation while promoting cardiomyocyte growth, whereas mitochondrial pyruvate carrier overexpression counteracts cardiac hypertrophic growth.^{13,14} Circumstantial evidence that this applies to humans is derived from patients with advanced HF undergoing LV assist device implantation.^{14,15} Compared with nonresponders, patients who recover LV function after mechanical unloading exhibit upregulation of enzymes involved in anabolic pathways branching from glycolysis, suggesting that LV recovery is associated with a shift from catabolism to anabolism.¹⁵

In HFrEF, neurohormonal activation stimulates lipolysis, consequently increasing ketone production by the liver.¹⁶ Evidence from both patients and animal models of HFrEF indicates increased myocardial uptake of ketones, driven by their elevated concentrations.^{6,8} It has been proposed that ketones represent a favorable substrate for the heart because their oxidation yields more ATP per consumed oxygen (expressed as the P/O ratio) compared with FA and glucose. Indeed, mice lacking the ability to oxidize β-hydroxybutyrate (β-OHB) in the heart develop systolic dysfunction during fasting and more severe dysfunction than wild-type mice after combined pressure overload and myocardial infarction.¹⁷ In humans, myocardial β-OHB levels are markedly elevated in end-stage HF.^{7,8} Furthermore, in patients with HFrEF, short-term infusion of β-OHB increased cardiac output without apparent impairment of myocardial energy efficiency,¹⁸ likely mediated by a combination of β-OHB-induced vasodilation and increased cardiac contractility.¹⁹ Collectively, these findings suggest that increased myocardial ketone metabolism may represent an adaptive response to preserve cardiac energetics in HFrEF.

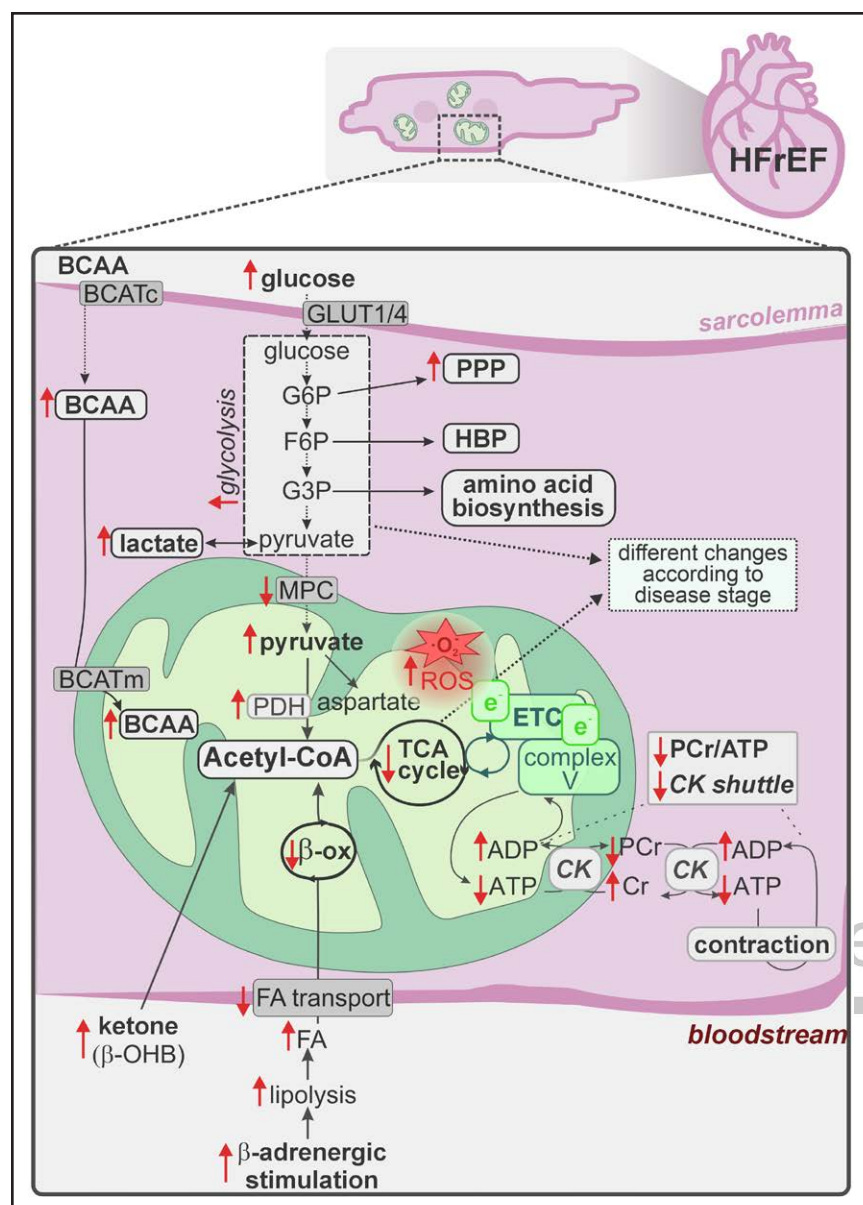


Figure 1. Cardiac substrate metabolism in heart failure with reduced ejection fraction (HFrEF).

In HFrEF, the majority of animal and human studies have reported a shift in substrate utilization toward glucose and lactate and away from fatty acids (FA). However, increased glucose uptake is not always associated with increased oxidation in mitochondria—the coupling between glycolysis and pyruvate oxidation might change according to the stage of heart failure (HF). Ketone utilization is increased as a result of elevated ketone levels in HFrEF. Elevated myocardial levels of branched-chain amino acids (BCAAs) were interpreted as a sign of impaired BCAA catabolism, but this notion has recently been challenged. Transfer of ATP to the site of its utilization in the cytosol via the creatine kinase (CK) shuttle is impaired, limiting cardiac inotropic reserve. Red arrows indicate changes in metabolite and enzyme levels in end-stage human DCM. β -OHB indicates β -hydroxybutyrate; β -ox, β -oxidation of fatty acid; BCAT, branched-chain amino acid aminotransferase; ETC, electron transport chain; HBP, hexosamine biosynthetic pathway; MPC, mitochondrial pyruvate carrier; PCr, phosphocreatine; PDH, pyruvate dehydrogenase; PPP, pentose phosphate pathway; ROS, reactive oxygen species; and TCA cycle, tricarboxylic acid cycle.

In the healthy heart, the contribution of amino acids to cardiac ATP production is marginal. Under fasting conditions, the human heart secretes most amino acids, indicative of protein breakdown.⁶ This negative nitrogen balance is exacerbated in patients with HFrEF, possibly reflecting an even greater degree of proteolysis. Interestingly, while levels of other amino acids are decreased, levels of branched-chain amino acids (BCAAs) are elevated in HF in humans and mouse models.^{7,20} Animal studies suggest that BCAA accumulation may contribute to HF by promoting mitochondrial dysfunction and oxidative stress. Conversely, enhancing BCAA catabolism has been shown to mitigate HF in mice subjected to transverse aortic constriction or myocardial infarction.²⁰ However, recent evidence proposes that BCAA oxidation is actually increased in the failing heart, and that the therapeutic benefits of stimulating BCAA catabolism derive

from reduced vascular resistance.²¹ Overall, alterations in amino acid metabolism, and particularly BCAA catabolism, remain an incompletely resolved area in HF research.

Changes in Cardiac Metabolism in HFpEF

Approximately 50% of patients with HF have a preserved ejection fraction. HFpEF arises from a complex interplay between cardiac-intrinsic abnormalities and systemic comorbidities, including diabetes, obesity, hypertension, and chronic kidney disease (CKD), resulting in distinct patient phenotypes. Together, cardiac-intrinsic alterations and systemic conditions impair cardiac relaxation and energetics via multiple mechanisms, which involve both alterations of the extracellular matrix and disruption of homeostatic and protective signaling pathways in cardiomyocytes.^{22,23}

Cardiometabolic HFpEF, arguably the most common HFpEF phenotype in Western countries,²⁴ develops in individuals with obesity and MetS and is characterized by myocardial FA overload, lipotoxicity and insulin resistance.²⁵ In contrast, in Eastern Europe and large parts of Asia, a lean hypertensive phenotype may predominate, with patients carrying a high burden of comorbidities despite being younger and leaner.^{26,27} Other HFpEF phenotypes are those of older individuals with advanced CKD and of patients affected by inflammatory/autoimmune diseases.²⁸ Although still preliminary, phenotype classifications have emerged from clinical trials and real-world cohorts²⁹ and are paralleled by diverse animal models, in which HFpEF hallmarks are reproduced with different hits.³⁰ Keeping this heterogeneity in mind is key when approaching cardiac energetics in HFpEF, especially when substrate preferences are investigated, as diverse metabolic backgrounds lead to different metabolic landscapes.

In HFpEF, mitochondrial dysfunction is a central contributor to impaired cardiac energetics. Ultrastructural abnormalities are consistently observed in myocardial samples from patients with HFpEF but are generally less severe than in HFrEF.³¹ This mitochondrial damage stems from several mechanisms, including excessive mitochondrial production of reactive oxygen species,³² altered mitochondrial dynamics,³¹ and suppressed mitophagy, as particularly observed in rodent models combining pressure and metabolic overload.³³ Depletion of nicotinamide adenine dinucleotide (NAD), an essential coenzyme for oxidative phosphorylation, further compromises mitochondrial function in HFpEF.³⁴

Data on FA utilization are conflicting between animal models and human studies. In a mouse model of cardiometabolic HFpEF induced by hypertension and obesity, oxidation of both palmitoyl-carnitine and pyruvate in isolated cardiac mitochondria is impaired, indicating intrinsic oxidative dysfunction. Yet, when comparing glucose and FA oxidation, an increased reliance on FA is observed at the organ level.^{33–35} In contrast, human HFpEF studies involving mainly obese subjects reveal reduced myocardial FA uptake without compensatory increases in ketone utilization, glycolysis, or BCAA metabolism^{36,37} (Figure 2). Transcriptomic and metabolomic analyses confirm downregulation of FAO-related genes and depletion of medium- and long-chain acylcarnitines in HFpEF hearts.^{36,38}

This discrepancy between species may reflect the differences between the 2 settings. While rodents typically model early-stage HF in young, male animals, human cohorts often reflect advanced disease in older, more often female, more heterogeneous, and comorbid populations. Also, preclinical analyses are performed measuring FA uptake and utilization in perfused hearts or isolated cardiac mitochondria, whereas human data are mainly based on indirect assessments from endomyocardial

biopsies of the right ventricle compared with unused explanted donor hearts as controls.³⁶

Glucose metabolism is also altered in HFpEF, albeit variably depending on the model or patient profile. In hypertensive or aging rodent models, increased glycolysis is observed without a matching increase in glucose oxidation.³⁹ In contrast, obese HFpEF models show reductions in both glucose uptake and oxidation, alongside increased FAO.^{33,35} Myocardial metabolomic and substrate uptake studies indicate that in human HFpEF the heart exhibits reduced glucose uptake and increased lactate extraction,³⁷ decreased levels of glycolytic intermediates (except pyruvate and lactate), and dysregulation of key metabolic enzymes and transporters, including upregulation of GLUT1 and PDH kinase 4, and downregulation of hexokinase and phosphofructokinase.^{7,38}

Unlike HFrEF, neither myocardial nor circulating β -OHB levels are elevated in patients with HFpEF, and ketone uptake remains unchanged relative to nonfailing hearts,³⁶ with reduced expression of β -OHB dehydrogenase 1, a key enzyme for ketone oxidation, in cardiometabolic/aging HFpEF animal models.^{33,35} This suggests that, in contrast to the compensatory increase in ketone utilization seen in HFrEF, HFpEF hearts do not enhance ketone oxidation.

Amino acid metabolism is also perturbed. As in HFrEF, proteolysis appears active in HFpEF, as evidenced by myocardial amino acid uptake and metabolomic profiling.^{36,37} Myocardial BCAA levels are higher in HFpEF than in HFrEF, whereas BCAA catabolites are reduced in both, suggesting that impaired BCAA catabolism represents a common feature across the HF spectrum.

Although observations in animal models and humans, are discordant, these defects reflect a metabolic phenotype marked by substrate inflexibility and inefficient ATP production, core features that may contribute to impair the lusitropic reserve and lead to exercise intolerance in HFpEF.

Energy Starvation Hypothesis Revisited

The long-standing energy starvation hypothesis posits that impaired myocardial energetics precede and contribute to contractile dysfunction in HFrEF. This concept is supported by *in vivo* by ³¹P magnetic resonance spectroscopy studies showing a reduced phosphocreatine (PCr)-to-ATP ratio, which strongly correlates with mortality in both ischemic and nonischemic HFrEF.⁴⁰ This has provided the rationale for therapeutic strategies aimed at improving oxygen efficiency by shifting substrate utilization from FA to glucose, which has a higher P/O ratio (2.58 versus 2.33). However, while this shift may enhance oxygen efficiency, it does not necessarily increase energetic efficiency in terms of total ATP yield: full oxidation of 1 mol of palmitate generates ≈ 105 to 110 mol of ATP, whereas 1 mol of glucose yields ≈ 30 to 32 mol of ATP.

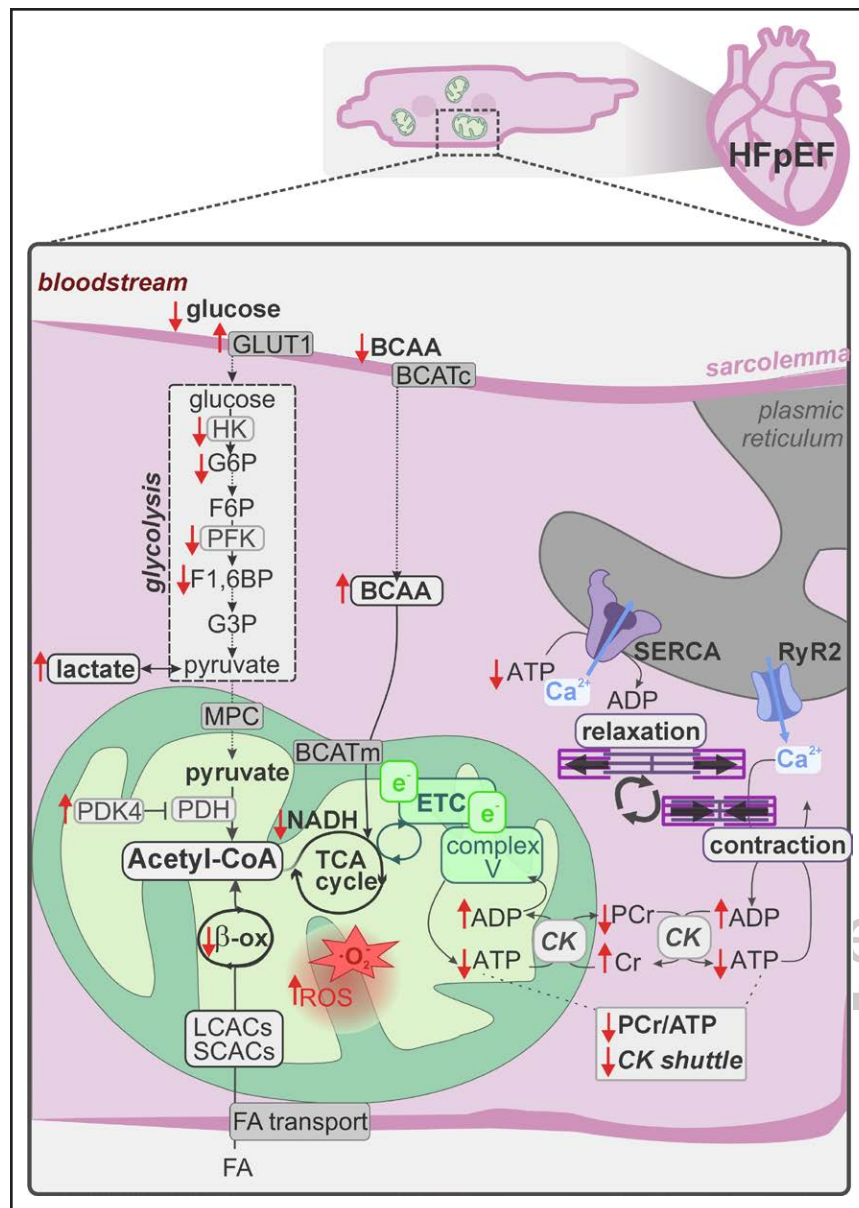


Figure 2. Cardiac substrate metabolism in heart failure with preserved ejection fraction (HFpEF).

In HFpEF, data on fatty acid (FA) utilization are conflicting between animal models and human studies, possibly reflecting differences between species and experimental settings. Reduced glucose uptake, decreased levels of glycolytic intermediates, and increased lactate extraction and utilization have been observed in human HFpEF hearts. Myocardial branched-chain amino acid (BCAA) levels are elevated in human HFpEF. Akin to heart failure with reduced ejection fraction (HFrEF), ATP transfer via the creatine kinase (CK) shuttle is impaired. Red arrows indicate changes in metabolite and enzyme levels in human HFpEF studies. β -ox indicates β -oxidation of fatty acid; BCAT, branched-chain amino acid aminotransferase; ETC, electron transport chain; HBP, hexosamine biosynthetic pathway; LCAC, long-chain acylcarnitine; MPC, mitochondrial pyruvate carrier; NADH, nicotinamide adenine dinucleotide (reduced form); PCr, phosphocreatine; PDK4, pyruvate dehydrogenase kinase 4; PDH, pyruvate dehydrogenase; PPP, pentose phosphate pathway; ROS, reactive oxygen species; RyR2, ryanodine receptor 2; SERCA, sarcoplasmic reticulum calcium ATPase; SCAC, short-chain acylcarnitine; and TCA cycle, tricarboxylic acid cycle.

In HFrEF, PCr levels decline early, whereas myocardial ATP concentrations remain relatively preserved until advanced stages of the disease,⁴¹ arguing against insufficient ATP production as a limiting factor for cardiac function at rest. Furthermore, reductions in CK flux exceed the decline in the PCr/ATP ratio,^{41,42} indicating that impaired transfer of high-energy phosphates from mitochondria to cytosolic ATPases – rather than insufficient ATP production per se—may represent the dominant defect. This limitation might become especially consequential during conditions of increased ATP turnover, such as exercise.⁴³

A similar energetic phenotype is observed in HFpEF, where reduced PCr/ATP ratios are present at rest and during exertion and correlate with impaired diastolic function and abnormal exercise hemodynamics.^{44,45} Notably, a $\approx 15\%$ reduction in the PCr/ATP ratio is also seen in

subjects with either obesity or diabetes without HF, suggesting that systemic metabolic perturbations impair myocardial energetics.^{46,47} In HFpEF, altered energy transfer may constrain myocardial relaxation by limiting sarcoplasmic reticulum calcium reuptake via SERCA, an enzyme that is particularly sensitive to elevated ADP levels due to its reliance on a steep energetic gradient.⁴⁸

Together, these findings indicate that while HFrEF and HFpEF exhibit distinct alterations in substrate metabolism, both share a final common pathway of energetic deficiency that does not reflect a primary ATP shortage but rather impaired energy transfer and reduced metabolic flexibility, ultimately limiting systolic and diastolic reserve. These observations suggest that therapeutic strategies should not be limited to increasing ATP production per se, but rather should aim to restore efficient energy transfer and metabolic flexibility, particularly under

stress conditions. Interventions capable of improving mitochondrial-cytosolic coupling or enhancing reserve capacity during exertion may therefore represent a more rational and clinically relevant target.

SYSTEMIC METABOLIC ALTERATIONS IN HF

Metabolic Syndrome and Metaflammation

MetS, defined by the cooccurrence of insulin resistance, central adiposity, dyslipidemia, and hypertension, shapes cardiac metabolism in HF, particularly HFpEF, through both direct and indirect mechanisms. Direct effects include elevated circulating free FAs (FFAs), insulin resistance leading to impaired glucose uptake and oxidation, increased oxidative stress, and activation of profibrotic and proinflammatory signaling pathways. Indirectly, MetS promotes coronary atherosclerosis, microvascular dysfunction, concentric hypertrophy (often driven by chronic hypertension), and increased deposition of epicardial AT, all of which can further impair myocardial perfusion and substrate utilization.

In addition, all components of MetS contribute to a state of chronic, low-grade systemic inflammation, referred to as metaflammation, which plays a role in HF pathogenesis.²³ Metaflammation arises at the interface of the immune and metabolic systems within energy-storing organs, particularly AT and the liver, where nutrient excess is sensed by resident immune cells and triggers sustained inflammatory signaling.⁴⁹ The following sections will explore how the altered crosstalk between metabolically active organs and the heart contributes to the pathophysiology of HF.

Interorgan Crosstalk: Heart and AT

Changes in AT significantly affect cardiac metabolism, particularly in obesity, insulin resistance, and type 2 diabetes. In HF, altered AT-derived signals, mediated by adipokines, lipids, extracellular vesicles (EVs), and inflammatory cytokines, contribute to metabolic remodeling in cardiomyocytes. Pathological AT remodeling in patients with MetS, also known as adiposopathy, is characterized by adipocyte hypertrophy, immune cell infiltration, fibrosis, capillary rarefaction, and mitochondrial dysfunction.⁵⁰ In humans, these changes shift the secretory profile toward increased inflammatory cytokines⁵¹ and a decreased adiponectin/leptin ratio.⁵² The inflammatory milieu impairs insulin-mediated suppression of hepatic lipogenesis and AT lipolysis, leading to elevated circulating FFAs and lipid redistribution to ectopic sites, including the heart (Figure 3).

Excess lipid supply to non-ATs, including the myocardium, can lead to the accumulation of complex lipid species such as ceramides, exerting deleterious effects

on cellular function. Clinical data indicate that elevated plasma ceramide levels are associated with increased cardiovascular risk,⁵³ although their primary source remains unclear. Increased ceramide levels, especially C16:0 and C18:0, have been described in the LV myocardium of animal models of HFpEF,³³ but data on myocardial ceramide content in HFrEF are inconsistent, at least partly because of differences in cause and sampling location.⁷

Visceral AT (VAT) is particularly relevant for its capacity to supply FFAs and proinflammatory adipokines. VAT undergoes marked remodeling in individuals with MetS and contributes to HF pathophysiology through both systemic and local mechanisms. In patients with HFpEF, VAT is increased and correlates with impaired LV function⁵⁴ and reduced exercise capacity.⁵⁵ In close anatomic proximity to the heart, the epicardial AT further exacerbates this maladaptive crosstalk. The epicardial AT is a metabolically active white fat depot that directly bathes the myocardium in FFAs, cytokines, and EVs, linking local AT with increased HF risk in clinical cohorts.⁵⁶

The AT functions as a dynamic endocrine organ, with adipokines mediating interorgan communication. Among these, leptin, whose circulating levels in humans reflect total fat mass and adipocyte size, plays a multifaceted role. Beyond its appetite-suppressing effects, leptin influences cardiometabolic pathways through the OB-R receptor and JAK2/STAT3 signaling in mice⁵⁷ and acutely promotes myocardial glucose utilization in ob/ob mice.⁵⁸ In obesity, however, chronic hyperleptinemia becomes maladaptive. Preclinical data indicate that the development of central leptin resistance promotes hyperphagia and further adiposity,⁵⁹ while leptin activates neurohormonal systems, including aldosterone secretion,⁶⁰ which may contribute to plasma volume expansion. This hemodynamic load can impose additional mechanical and energetic stress on the heart that may translate into oxidative stress and maladaptive cardiac remodeling via activation of redox-sensitive kinases (such as the Ca²⁺/calmodulin-dependent protein kinase II [CaMKII]).⁶¹ In addition, leptin secreted by epicardial AT in rats may act locally on the myocardium through protein kinase C/NADPH oxidase signaling, thereby amplifying oxidative stress and mitochondrial dysfunction.⁶² Altogether, leptin exemplifies how the AT may contribute to HFpEF pathogenesis through both systemic endocrine actions and local paracrine effects.

In contrast, adiponectin is the most abundant adipokine in lean individuals and plays a protective role in cardiac metabolism. In humans, its levels decline with increasing visceral adiposity,⁶³ and this suppression is linked to metabolic dysregulation in HF. In rabbits, adiponectin enhances cardiac FAO, improves insulin sensitivity, and supports mitochondrial biogenesis and function.⁶⁴ Preclinical data in rodents indicate that adiponectin activates

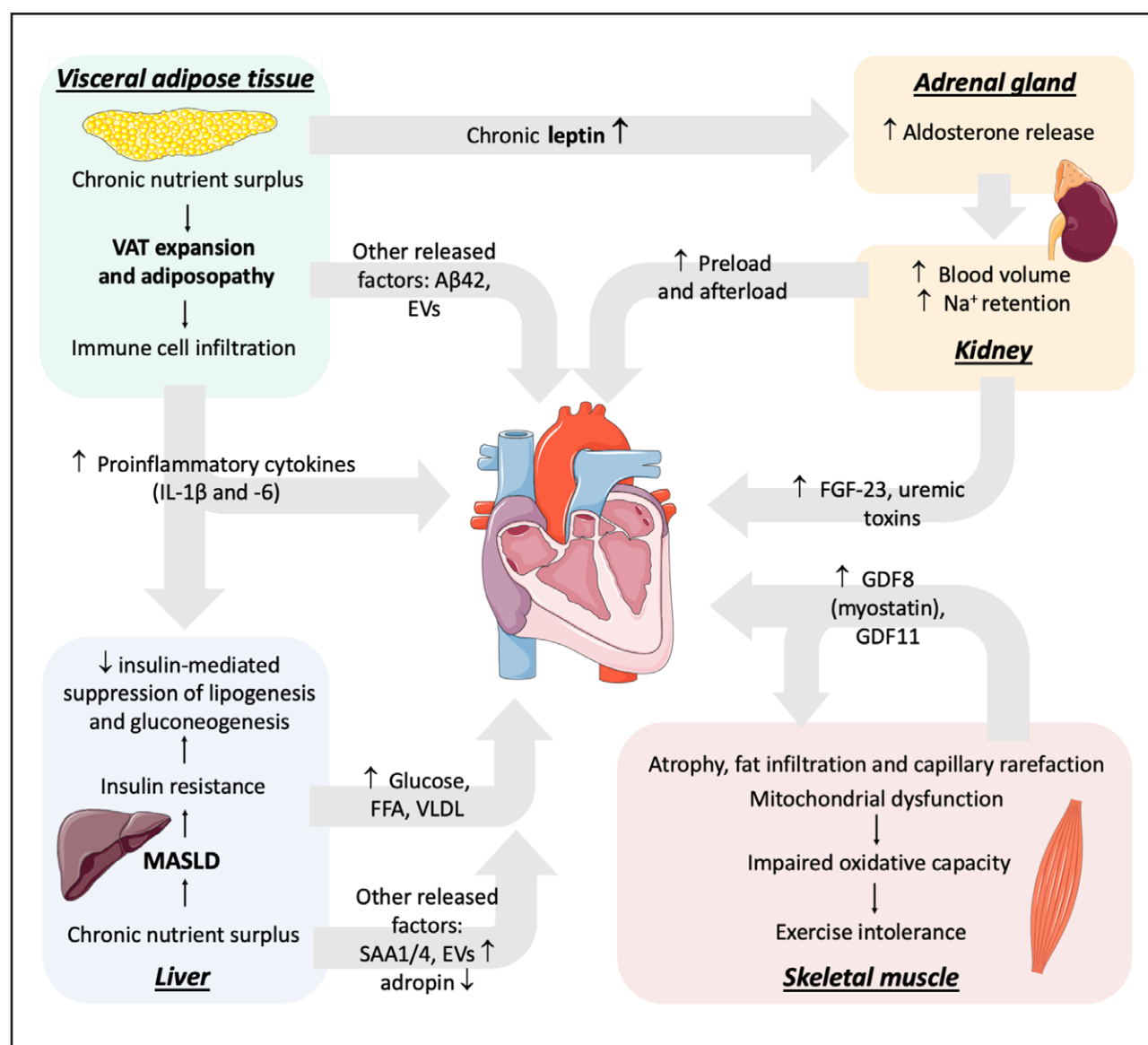


Figure 3. Interorgan crosstalk in cardiometabolic heart failure with preserved ejection fraction.

The figure illustrates the major alterations occurring in obesity and metabolic syndrome across metabolically active organs and tissues, and how these systemic changes impact the heart and contribute to the pathogenesis of HFpEF via released factors. Aβ42 indicates amyloid beta 42; EV, extracellular vesicle; FFA, free fatty acid; FGF23, fibroblast growth factor-23; MASLD, metabolic dysfunction-associated steatotic liver disease; SAA1/4, serum amyloid A1/4; VAT, visceral adipose tissue; and VLDL, very-low-density lipoprotein. Images of adipose tissue, liver, kidney, skeletal muscle, and heart are provided by Servier Medical Art (<https://smart.servier.com/>), licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).

key cytoprotective pathways in cardiomyocytes, such as SIRT1, PGC-1α, and PPARα, thereby mitigating oxidative stress and suppressing maladaptive remodeling.⁶⁵

This hormonal communication between AT and the heart is further complemented by the transfer of genetic material via EVs. In mouse models of diabetes, adipocyte-derived EVs carrying miR-130b-3p impair mitochondrial respiration in cardiomyocytes by downregulating PGC-1α, increasing reactive oxygen species, and activating apoptotic pathways. These effects worsen ischemia/reperfusion injury but can be reversed by blocking miR-130b-3p or overexpressing PGC-1α.⁶⁶ Similarly,

PPARγ activation in adipocytes promotes EV secretion containing miR-200a, which promotes pathological cardiac hypertrophy by activating mechanistic target of rapamycin signaling. This maladaptive signaling is prevented in mice lacking PPARγ specifically in AT.⁶⁷

Obesity has also been shown to induce adipose expression of amyloidogenic proteins, thereby increasing circulating amyloid beta 42, which accumulates in cardiac mitochondria in mice.⁶⁸ Amyloid beta 42 inhibits complex I, induces downregulation of tricarboxylic acid cycle enzymes and PDH, and promotes metabolic inflexibility. In lean mice, exogenous amyloid beta 42 reproduces

HF features, while an amyloid beta 42-neutralizing antibody reverses these effects, underscoring its pathogenic potential.

There is clinical evidence that supports these mechanisms, although a causal relationship is difficult to prove. In obese patients undergoing bariatric surgery, reductions in VAT and epicardial AT mass are paralleled by improved myocardial energetics and by structural and functional cardiac improvements.⁶⁹ These findings support the reversibility of metabolic derangements driven by AT and reinforce the therapeutic potential of targeting AT-cardiac crosstalk in HF. Importantly, the reversibility of these alterations with weight loss and metabolic interventions indicates that adipose-driven cardiac remodeling is not merely associative but modifiable. Therapies that reduce visceral and epicardial adiposity or attenuate metaflammation may therefore directly influence myocardial energetics and disease progression in cardiometabolic HFpEF.

Interorgan Crosstalk: Heart and Liver

Liver diseases can impair cardiac function through multiple mechanisms, including altered hemodynamics, vascular tone, circulating inflammatory mediators, and disrupted signaling pathways.⁷⁰ Especially, among patients with obesity or type 2 diabetes, metabolic dysfunction-associated steatotic liver disease (MASLD) affects myocardial substrate imbalance and energetic impairment. Formerly termed nonalcoholic fatty liver disease, MASLD comprises a spectrum of liver disorders characterized by excess hepatic fat accumulation independent of significant alcohol intake, and is highly prevalent among patients with HFpEF.⁷¹

In MASLD, hepatic insulin resistance promotes aberrant gluconeogenesis and de novo lipogenesis, thereby increasing hepatic output of glucose, triglycerides, and very-low-density lipoproteins, which in turn elevates circulating FFAs and induces lipotoxicity in target organs. In cardiometabolic HFpEF, where FFA oversupply and lipotoxicity are key pathogenic mechanisms, these liver-derived lipid species likely represent major contributors. Imaging studies support this link: patients with MASLD exhibit reduced myocardial glucose uptake on ¹⁸F-FDG PET, which correlates with diastolic function.⁷² Clinical data further indicate that MASLD, particularly hepatic insulin resistance and liver fibrosis, is associated with reduced myocardial mechanoenergetic efficiency, an echocardiographic index of the heart's ability to convert chemical energy into mechanical work.⁷³ Beyond its role in lipid and glucose metabolism, the liver also regulates ketone production, which provides an alternative substrate in HFrEF.⁸ However, MASLD impairs hepatic ketogenesis due to mitochondrial dysfunction and suppression of FAO,⁷⁴ potentially limiting this adaptive fuel source.

The liver also communicates with the myocardium through hepatokines secreted in response to metabolic or inflammatory cues. In preclinical models of HFpEF, FGF21 ameliorated diastolic function by activating AMPK signaling and adiponectin release, consequently promoting FAO and restoring PDH flux.⁷⁵ FGF21 signaling has also been implicated in the cardioprotective effects of exercise in humans and mice with diabetes.⁷⁶ Another hepatokine, adropin, is downregulated in MASLD and obesity in humans, and its deficiency is linked to reduced myocardial glucose oxidation and impaired diastolic performance in rodents.⁷⁷ In HF mouse models, adropin supplementation activates PDH, increases glucose oxidation, and enhances mitochondrial flexibility, suggesting a beneficial role in regulating cardiac fuel selection.⁷⁸

Liver-derived inflammatory and hemostatic mediators may further shape the liver-heart axis in cardiometabolic disease. Systems biology data indicate that serum amyloid A1/4 (SAA1/4), strongly induced in MASLD, plays a role in amplifying systemic inflammation and extracellular matrix deposition.⁷⁹ By contrast, in preclinical models, hepatocyte-derived FXI engages cardiac BMP-SMAD1/5 signaling to restrain inflammation and fibrosis, and lower circulating FXI levels have been observed in HFpEF cohorts, consistent with a loss of this compensatory signal.^{80,81}

The liver also influences the heart via EVs, which transport lipids, proteins, and regulatory RNAs. In patients with MASLD, hepatocyte-derived EVs become enriched in miR-122, a liver-specific microRNA targeting mitochondrial regulators.⁸²

In summary, the liver actively contributes to myocardial metabolic remodeling in HF, particularly cardiometabolic HFpEF, through diverse and interconnected pathways, including hepatic insulin resistance, lipid overflow, and impaired ketogenesis, which disrupt cardiac substrate homeostasis and promote mitochondrial dysfunction. These findings raise the possibility that targeting hepatic insulin resistance and steatosis may exert cardiovascular benefits beyond glycemic control. Whether pharmacological or lifestyle strategies aimed at improving liver metabolism translate into measurable improvements in cardiac energetics and outcomes in HFpEF warrants further investigation.

Interorgan Crosstalk: Heart and Skeletal Muscle

Reduced exercise tolerance is a hallmark of HFpEF and results from both central and peripheral abnormalities. In addition to an impaired increase in cardiac output due to chronotropic incompetence, SM dysfunction is a major determinant of reduced exercise capacity. In HFpEF, exercise-induced vasodilation in SM is blunted due to endothelial dysfunction and capillary rarefaction, limiting oxygen delivery. These vascular deficits are

accompanied by reduced muscle mass, a shift toward glycolytic type II fibers, and increased intramuscular fat infiltration (myosteatosis), all of which impair oxygen extraction and reduce muscular efficiency during physical activity.⁸³ Furthermore, SM mitochondria from patients with HFpEF exhibit diminished oxidative capacity compared with healthy controls, a deficit that correlates with reduced exercise performance.⁸⁴ The mechanisms driving SM remodeling in HFpEF are not fully understood. While excess AT has been associated with mitochondrial dysfunction and lower mitochondrial density,⁸⁵ evidence from both human studies and animal models of HFpEF suggests that additional systemic and local factors contribute to SM pathology.⁸⁴

Both the heart and SM produce and release signaling molecules, termed myokines, which affect the metabolic program of myocytes. Among these, GDF8 (growth differentiation factor 8, also known as myostatin) and its closely related paralog GDF11 promote a catabolic switch in both heart and SM. GDF8 expression is elevated in SM biopsies from patients with HFpEF⁸⁶ and is even higher in patients with HFpEF.⁸⁷ GDF8 silencing prevented SM wasting in HFpEF mice,⁸⁸ and GDF8-null mice exhibited reduced adiposity, possibly resulting from glucose diversion away from AT.⁸⁹ Although GDF11 was initially reported to reverse age-related cardiac and skeletal muscle alterations,⁹⁰ subsequent studies demonstrated that GDF11 activates the same signaling pathway as GDF8 and may contribute to age-related sarcopenia.⁹¹ Both GDF8 and GDF11 signal through the activin type II receptor, whose expression is upregulated in myocardial tissue from patients with end-stage HF.⁹² Overall, although the functional roles of GDF8 and GDF11 in human HF remain incompletely understood, preclinical evidence suggests that they act as negative regulators of SM mass and may contribute to the muscle-wasting phenotype observed in HF. It is noteworthy that drugs inhibiting the activity of the activin receptor type II, which binds GDF8 and GDF11, are already available for the treatment of pulmonary arterial hypertension and that, in animal models of HF, blockade of this receptor preserved SERCA levels and cardiac function.⁹³

Exercise training improves functional capacity and quality of life in patients with HFpEF.⁹⁴ Human studies indicate that these benefits are driven by peripheral adaptations within SM rather than by changes in cardiac function.⁹⁵ In a randomized controlled trial comparing different exercise modalities in HFpEF, high-intensity interval training induced greater upregulation of PGC-1 α and electron transport chain complexes in SM biopsies than moderate continuous training, although both interventions similarly improved peak oxygen consumption.⁹⁶ Consistently, in the obese hypertensive mouse model of HFpEF, 5 weeks of voluntary wheel running rescued defects in FA and BCAA oxidation and restored exercise capacity without affecting cardiac function.⁹⁷ Exercise

also reshapes the myokine milieu. Musclin, released from SM in response to physical activity, enhances endurance in mice by promoting mitochondrial biogenesis⁹⁸ and prevents myosteatosis by suppressing proliferation of fibro-adipogenic progenitors.⁹⁹ Irisin is another exercise-responsive myokine that acutely increases glucose uptake and oxidative metabolism in mice. However, whether circulating irisin levels predict cardiovascular risk remains uncertain, as studies have yielded conflicting results.¹⁰⁰ Importantly, both musclin and irisin also influence AT remodeling and thermogenesis.^{101,102}

Altogether, these findings highlight SM abnormalities as critical determinants of exercise intolerance in HFpEF. Physical exercise can rescue SM alterations by restoring mitochondrial function, substrate utilization, and systemic metabolic crosstalk, ultimately improving exercise performance in HFpEF. The improvement in functional capacity with exercise training underscores that peripheral metabolic dysfunction is a modifiable determinant of HFpEF severity. Future trials should determine whether targeting SM metabolism through exercise, pharmacological myostatin inhibition, or metabolic supplementation can translate into improvements in HF symptoms and events.

Chronic Kidney Disease

Patients with CKD have an increased risk of HF.¹⁰³ CKD-related HFpEF, or cardiorenal HFpEF, is characterized by myocardial fibrosis and LV hypertrophy due to multiple factors. Activation of the renin-angiotensin-aldosterone system plays a crucial role, promoting cardiac hypertrophy through increased vascular resistance, volume retention, and angiotensin II-mediated myocyte growth.¹⁰⁴ Additional contributors include plasma volume expansion, hormonal imbalances affecting the vitamin D-parathyroid hormone-FGF23 axis, hyperphosphatemia, and uremic toxins, all of which contribute to LV diastolic dysfunction.²⁸

The effects of CKD on cardiac metabolism in the context of cardiorenal HFpEF have only recently been addressed in rodent models.¹⁰⁴ Nephrectomized mice exhibit decreased cardiac FAO due to downregulation of the key transcription factor PGC-1 α .¹⁰⁵ FAO downregulation was observed also in rats subjected to nephrectomy.¹⁰⁶ Additional mechanisms are likely involved, including alterations in cardiomyocyte Ca²⁺ handling and systemic inflammation related to CKD.¹⁰⁷ However, developing accurate animal models of cardiorenal HFpEF and uremic cardiomyopathy presents challenges. Mice do not develop cardiac hypertrophy or fibrosis even with moderate-to-severe CKD,¹⁰⁸ and the impairment in FAO in CKD models may be independent of the occurrence of conduit arterial disease and LV hypertrophy.¹⁰⁹ For a more detailed overview of the impact of CKD on cardiac metabolism, in particular in the context of HF, we refer the reader to a recent excellent review.¹⁰⁴

THERAPEUTIC IMPLICATIONS AND FUTURE DIRECTIONS

Agents Directly Targeting Myocardial Substrate Utilization

Preclinical HFrEF models suggest that shifting fuel use toward glucose utilization or away from FAO can improve cardiac efficiency and function.¹¹⁰ Translation to humans, however, remains limited. The PDH kinase inhibitor dichloroacetate, which stimulates glucose oxidation by activating PDH, improved LV mechanical efficiency in patients with advanced HF and NYHA class III to IV symptoms.¹¹¹ Etomoxir and perhexiline inhibit FA import into mitochondria, but their use is limited by a narrow therapeutic window and hepatotoxicity.^{112,113} The FAO inhibitor trimetazidine, also used as an antianginal agent, improved functional capacity and HF symptoms in several small clinical trials with heterogeneous designs and

small sample sizes. More recently, the novel FAO inhibitor ninafostat improved cardiac energetics and diastolic function in obese patients with type 2 diabetes.¹¹⁴ Overall, despite encouraging preclinical and early clinical data, the effect of substrate modulation on HF trajectory in humans remains unproven.

Sodium-Glucose Cotransporter 2 Inhibitors

The benefits of sodium-glucose cotransporter 2 (SGLT2) inhibitors in HF have also been related to effects on cardiac and systemic metabolism. One of the earliest hypotheses postulated that SGLT2 inhibitors enhance myocardial ketone utilization (Figure 4). This notion was supported by clinical observations in fasting diabetic patients treated with SGLT2 inhibitors, who exhibited increased circulating ketones, and by the theoretical advantage of ketones as an energy substrate, given their higher P/O ratio compared with glucose or FA.¹¹⁵

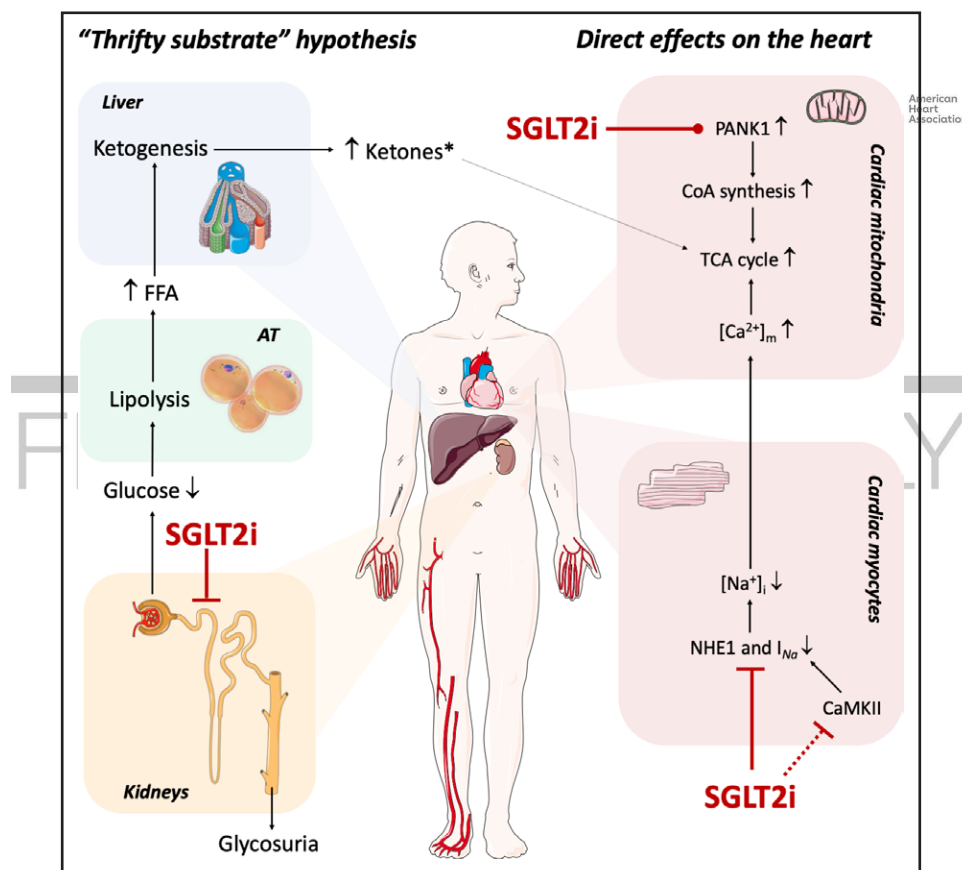


Figure 4. Proposed metabolic mechanisms of cardioprotection mediated by sodium-glucose cotransporter 2 (SGLT2) inhibitors.

It was initially proposed that SGLT2 inhibitors exert their cardioprotective effects by enhancing myocardial ketone utilization (thrifty substrate hypothesis, **left**), but subsequent investigations, including human trials, have shown that the effects of SGLT2 inhibitors on circulating ketone levels in nondiabetic individuals are modest or null. Preclinical evidence suggests that SGLT2 inhibitors exert direct effects on cardiomyocytes (**right**). These include activation of PANK1 (pantothenate kinase 1), the rate-limiting enzyme of coenzyme A (CoA) biosynthesis, and inhibition of sarcolemmal sodium currents, including the sodium/proton exchanger (NHE) and late sodium current (I_{Na}), which, by lowering intracellular sodium levels ($[Na^+]_i$), could lead to an increase in mitochondrial calcium ($[Ca^{2+}]_m$), thereby restoring the calcium-dependent stimulation of tricarboxylic acid (TCA) cycle enzymes. * Relevant only for diabetic patients. Black arrows indicate changes induced by SGLT2 inhibitors. Red arrows indicate direct effects of SGLT2 inhibitors. AT indicates adipose tissue; CaMKII, calcium/calmodulin-dependent kinase II; and FFA, free fatty acid. Images of the human body, nephron, adipocytes, liver lobule, cardiac myocyte, and mitochondrion were adapted from Servier Medical Art (<https://smart.servier.com/>), licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).

Experimental evidence further supported this model. In a swine model of HFrEF due to anterior myocardial infarction, empagliflozin enhanced myocardial ketone metabolism, increased ATP production, and attenuated LV remodeling.¹¹⁶ However, SGLT2 inhibitors also induced improvements in glucose oxidation and FAO, suggesting a broader enhancement of cardiac metabolic flexibility.¹¹⁶ Furthermore, human mechanistic studies have challenged this ketone-centric model. In the DEFINE-HF trial, increases in circulating ketones in nondiabetic patients with HFrEF receiving SGLT2 inhibitors were modest,¹¹⁷ and in the EMPA-VISION trial, empagliflozin did not alter the cardiac PCr/ATP ratio or β -OHB levels in patients with HFrEF or HFpEF compared with placebo.¹¹⁸

Alternative and potentially complementary mechanisms have emerged, accompanied by increasing evidence of a direct effect of SGLT2 inhibitors on cardiomyocytes, despite the absence of SGLT2 expression in cardiac tissue. SGLT2 inhibitors may induce a metabolic milieu resembling fasting, thereby activating AMPK- and hypoxia-related signaling pathways that promote autophagy and the clearance of toxic lipid intermediates.¹¹⁹ Although conceptually appealing, this model is primarily based on results in vitro and still awaits direct experimental validation. Furthermore, empagliflozin has been shown to inhibit key sodium (Na^+) transport mechanisms in cardiomyocytes, including the Na^+/H^+ exchanger 1¹²⁰ and the late Na^+ current (I_{NaL}) via voltage-gated Na^+ channels.¹²¹ These effects on ionic homeostasis may attenuate pathological Ca^{2+} overload, improve mitochondrial function, and contribute to the observed improvements in contractility and energetics.¹²² A recent ex vivo study demonstrated that empagliflozin improves oxidative metabolism in human myocardial samples, along with cardiomyocyte contraction and relaxation in vitro. This effect was linked to activation of PANK1 (pantothenate kinase 1), the rate-limiting enzyme in the synthesis of coenzyme A from vitamin B5, implicating coenzyme A availability as a novel target in SGLT2 inhibitor-mediated metabolic remodeling¹²³ (Figure 4).

In conclusion, the cardioprotective effects of SGLT2 inhibitors likely arise from a combination of systemic and myocardial actions, including a modest modulation of ketone metabolism and, thereby, myocardial bioenergetics and a direct enhancement of cardiac energetics and contractility. Importantly, their clinical efficacy suggests that targeting metabolic and energetic stress can alter HF outcomes.

Glucagon-Like Peptide-1 Receptor Agonists

GLP-1RA (glucagon-like peptide-1 receptor agonists) represent a new therapeutic option for obese, mainly cardiometabolic, HFpEF and HF with mildly reduced ejection fraction.¹²⁴ In addition, tirzepatide, a dual GLP-1

and GIP (glucose-dependent insulintropic polypeptide) receptor agonist, reduced the risk of worsening HF events in obese patients with HFpEF in the SUMMIT trial.¹²⁵ Although SUMMIT had a relatively short follow-up and a limited number of HF events, it provided the first signal that incretin-based therapies may modify the clinical trajectory of cardiometabolic HFpEF.

GLP-1 and GIP receptor agonists mimic the physiological actions of incretin hormones, acting primarily on gastric emptying, pancreatic islets, and the central nervous system to improve glucose homeostasis, reduce visceral adiposity, and lower body weight.¹²⁶ Importantly, weight loss per se, achieved through lifestyle interventions or bariatric surgery, is associated with improved exercise capacity and reduced risk of incident HF events, highlighting excess adiposity as a driver of cardiometabolic HFpEF.¹²⁷ Mechanistically, weight loss is associated with a reduction in circulating blood volume, which may lower LV end-diastolic filling pressures and, in the SUMMIT trial, correlated with improvements in both KCCQ score and 6-minute walk distance.¹²⁸ Therefore, a key question is whether GLP-1RAs exert cardiovascular effects beyond those expected from weight reduction alone. In a preclinical HFpEF model combining advanced age, female sex, obesity, and type 2 diabetes, semaglutide improved cardiac function more than matched caloric restriction, implicating additional mechanisms related to improved endothelial function and restoration of protective immune responses within VAT.¹²⁹ Indeed, GLP-1RAs also exert antiinflammatory effects, partly through modulation of gut intraepithelial lymphocytes¹³⁰ and by reducing inflamed VAT. In SUMMIT, the antiinflammatory activity of tirzepatide, reflected by decreased C-reactive protein, emerged later than the hemodynamic effects and showed only a modest correlation with functional improvement.¹²⁸ Preclinical studies further suggest cardioprotective actions beyond weight loss, as semaglutide lowers elevated cytosolic Ca^{2+} transients in cardiomyocytes from obese rats,¹³¹ enhances mitochondrial fusion,¹³² and augments nitric oxide signaling in HFpEF mouse models.¹³³

The GLP-1 receptor is detectable in the human sinoatrial node, which explains the modest increase in heart rate seen with these agents, but its presence and functional relevance in other cardiac regions are unclear. Indeed, GLP-1RAs retain cardioprotective effects even when the cardiomyocyte GLP-1 receptor is selectively deleted in mice.¹³⁴ However, in human cardiomyocytes from patients with HFrEF or aortic stenosis (with a HFpEF phenotype), semaglutide reduced late I_{Na} , decreased diastolic Ca^{2+} leak from the sarcoplasmic reticulum and improved systolic Ca^{2+} transients and contractility via GLP-1 receptor activation, effects that were comparable to CaMKII inhibition.¹³⁵

Overall, additional studies are needed to determine to which extent GLP-1- and GIP-based therapies modulate

systemic or cardiac metabolism, as well as cardiac Ca^{2+} and Na^+ handling.

Supplementation With NAD Precursors

NAD is a central coenzyme in oxidative metabolism, functioning as an electron carrier in key dehydrogenase reactions of glycolysis and the tricarboxylic acid cycle. Beyond its role in redox reactions, NAD serves as a cosubstrate for enzymes involved in posttranslational and epigenetic regulation, including sirtuins, PARPs (poly[ADP-ribose] polymerases), and CD38. These enzymes irreversibly degrade NAD into nicotinamide and ADP-ribose, linking NAD availability to cellular functions other than oxidative metabolism, such as DNA repair, Ca^{2+} signaling, and protein deacetylation. Cellular NAD content is determined by the balance between its enzymatic consumption and its synthesis, either *de novo* or via the salvage pathway.

Strategies to restore NAD levels, either through supplementation with NAD precursors or stimulation of the salvage pathway, have improved cardiac function in animal models of HF.^{34,136,137} Among available precursors, nicotinamide mononucleotide and nicotinamide riboside have been tested in small clinical studies.^{138,139} Myocardial NAD depletion has been documented in animal models of HFrEF and in both animal models and patients with HFpEF,^{137,140} but not consistently in human HFrEF.⁷ A possible mechanism underlying NAD depletion is reduced activity of nicotinamide phosphoribosyltransferase, the rate-limiting enzyme of the salvage pathway, which is downregulated in myocardial samples from patients with both HFrEF and HFpEF.^{34,140} In addition, certain myocardial stressors may enhance the activity of NAD-consuming enzymes, further lowering NAD availability.¹⁴¹

The mechanisms mediating the benefits of NAD supplementation remain incompletely defined. Restoring NAD levels may enhance sirtuin activity and thereby promote protein deacetylation, which has been suggested to be protective, particularly for mitochondrial proteins.^{136,142} However, improvements in cardiac function in the absence of major changes in the acetylome have also been observed.¹⁴³ In a cardiometabolic HFpEF model, the benefit of nicotinamide mononucleotide supplementation was attributed to restoration of autophagy.¹⁴⁴

The metabolic benefits of NAD supplementation may extend beyond the heart. In preclinical models, supplementation with NAD precursors improves SM oxidative capacity during aging and in the setting of diet-induced obesity.^{145,146} In humans, a 10-week administration of nicotinamide mononucleotide improved SM insulin sensitivity in prediabetic women.¹³⁸ However, the causal role of NAD depletion in SM dysfunction remains uncertain, as mice with an $\approx 85\%$ reduction in muscle NAD levels due to genetic silencing of the rate-limiting enzyme of the salvage pathway maintain largely

preserved SM structure and function.¹⁴⁷ Therefore, while augmenting NAD availability may enhance SM metabolic flexibility under conditions of metabolic stress, its necessity for basal muscle function and its therapeutic relevance in HF require further investigation.

Altogether, the mechanisms underlying NAD depletion and the extent of NAD deficiency remain incompletely defined, particularly in human HFrEF, as do the pathways mediating the benefits of NAD repletion. Evidence for myocardial NAD depletion is more consistent in cardiometabolic HFpEF, where impaired autophagy may render the myocardium especially responsive to NAD-targeting strategies.

CONCLUSIONS

Alterations in cardiac and systemic metabolism are integral components of the HF syndrome. In HFrEF, intrinsic myocardial energetic defects interact with neurohormonal activation, whereas in HFpEF, metabolic stress imposed by comorbidities shapes a multi-system disorder. Metabolic alterations in HF with mildly reduced ejection fraction remain poorly characterized. Recent clinical evidence demonstrates that modifying metabolic stress through SGLT2 inhibitors, GLP-1RAs, weight reduction, and physical exercise can improve symptoms and outcomes across the HF spectrum. These observations establish metabolic remodeling as a therapeutically actionable axis in HF. The next challenge lies in identifying phenotype-specific metabolic alterations that can be targeted to alter disease trajectory, thereby translating mechanistic insights into precision metabolic therapies.

ARTICLE INFORMATION

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